Sanketh
tumul

Gene Cloning

DNA CLONING

A method for identifying and purifying a particular DNA fragment (clone) of interest from a complex mixture of DNA fragments, and then producing large numbers of the fragment (clone) of interest.

Why Clone DNA?

- A particular gene can be isolated and its nucleotide sequence determined
- Control sequences of DNA can be identified & analyzed
- Protein/enzyme/RNA function can be investigated
- Mutations can be identified, e.g. gene defects related to specific diseases
- Organisms can be 'engineered' for specific purposes, e.g. insulin production, insect resistance, etc.

Why synthesize human insulin?

- Patients' immune systems do not produce antibodies against human insulin as they do with bovine or porcine insulin
- Projected decline in the production of animal-derived insulin
- Need for a more reliable and sustainable method of obtaining the product

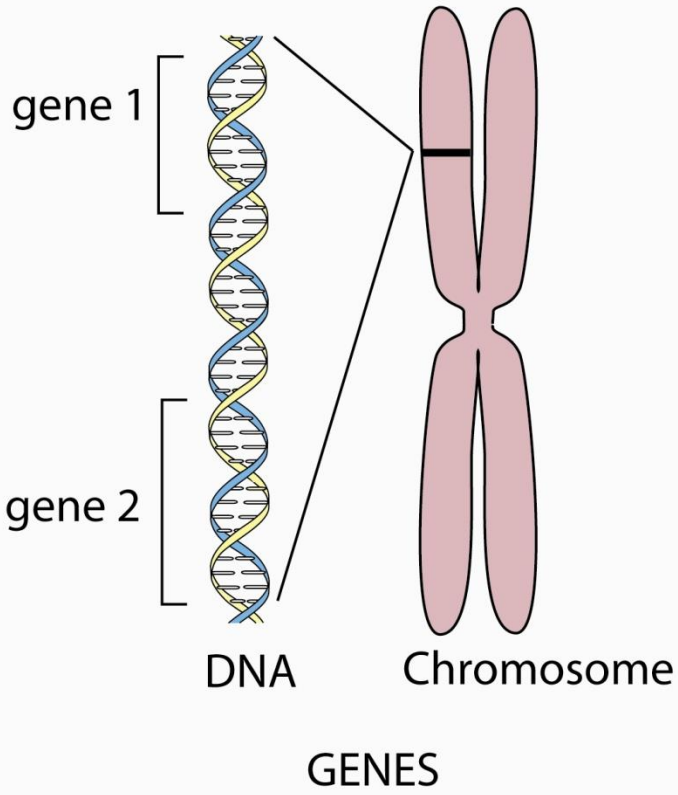
Cloning - a definition

- An individual grown from a single somatic cell of its parent & genetically identical to it

- *Clone: a collection of molecules or cells, all identical to an original molecule or cell*

Gene cloning

- When DNA is extracted from an organism, all its genes are obtained
- In gene (DNA) cloning a particular gene is copied (cloned)



Sources of DNA for Cloning

- 1) Chromosomal DNA
- 2) RNA converted to cDNA
- 3) PCR-amplified DNA

STEPS IN CLONING

Step 1: Identification & Isolation of Gene of interest or DNA fragment to be cloned.

Step 2: Insertion of this isolated gene in a suitable vector .

Step 3: Introduction of this vector into a suitable organism/ cell called host (transformation) .

Step 4: Selection of the transformed host cell.

Step 5: Multiplication or expression of the introduced gene in the host.

PCR-amplified DNA

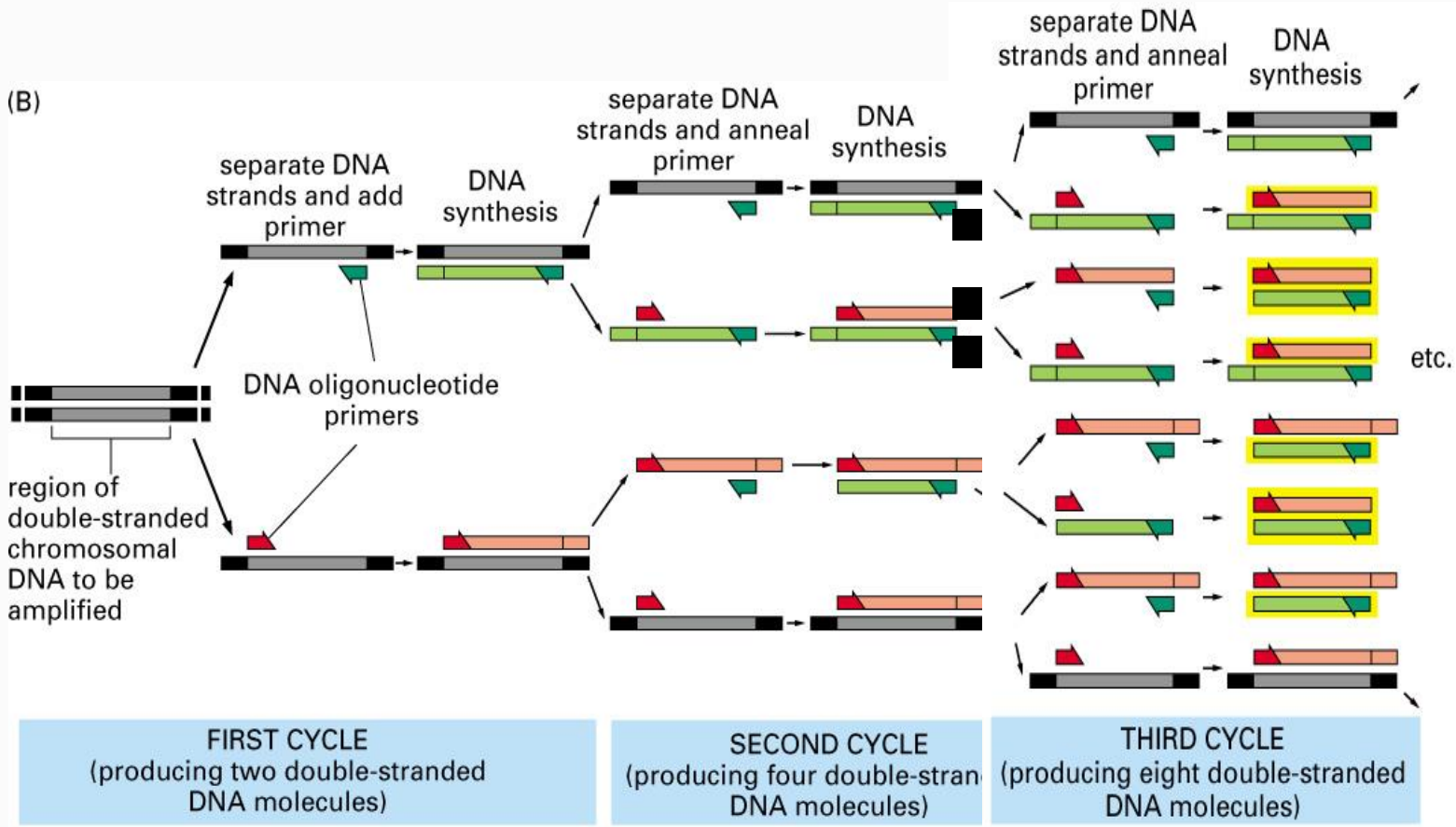


Figure 8–39 part 2 of 3. Molecular Biology of the Cell, 4th Edition.

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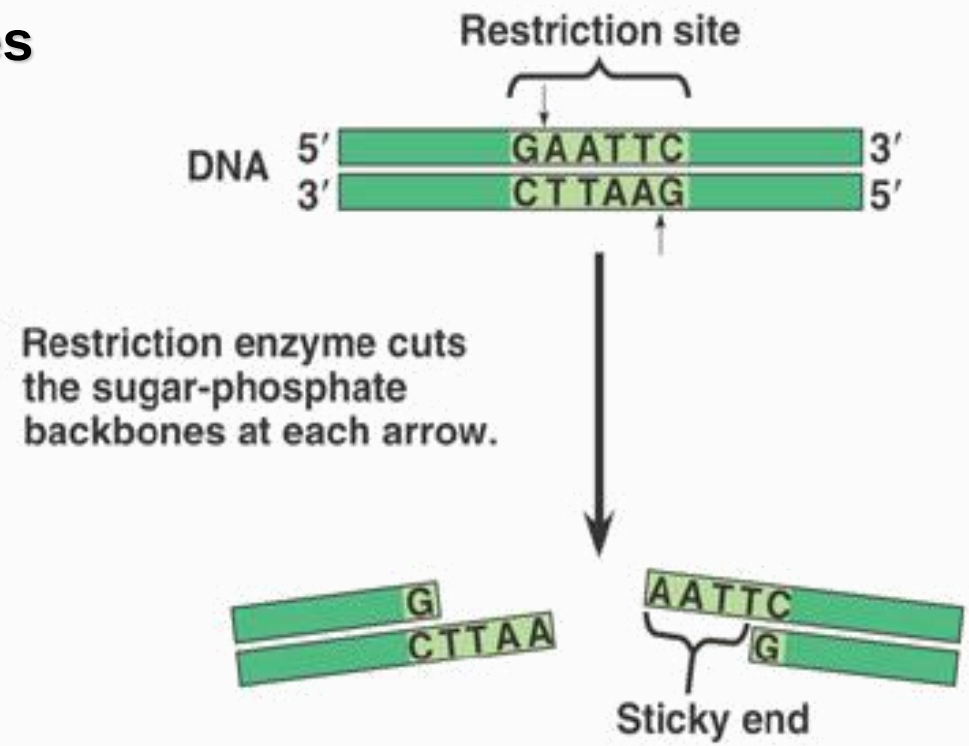
Cloning Tools

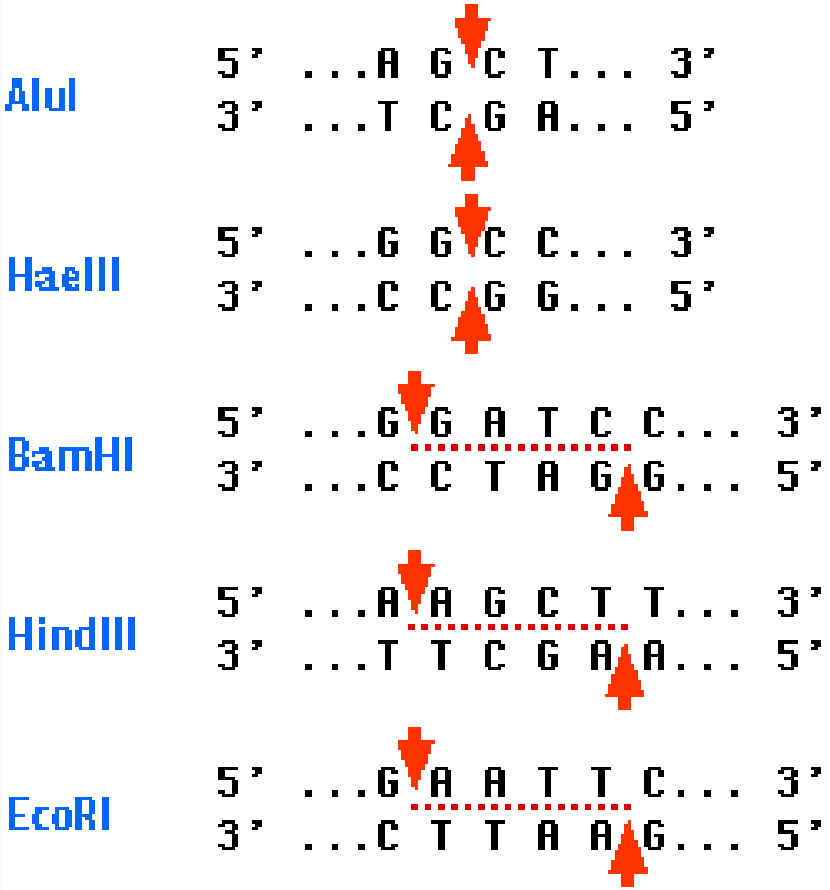
- Restriction endonucleases
- Ligase
- Vectors
- Host
- Methods for introducing DNA into a host cell

Cutting DNA

- **Restriction endonucleases (restriction enzymes)**
  - **sticky ends**
  - **blunt ends**
- **Nomenclature**
  - ***Eco*RI**
  - ***E* = genus (*Escherichia*)**
  - ***co* = species (*coli*)**
  - **R = strain**
  - **I = # of enzyme**



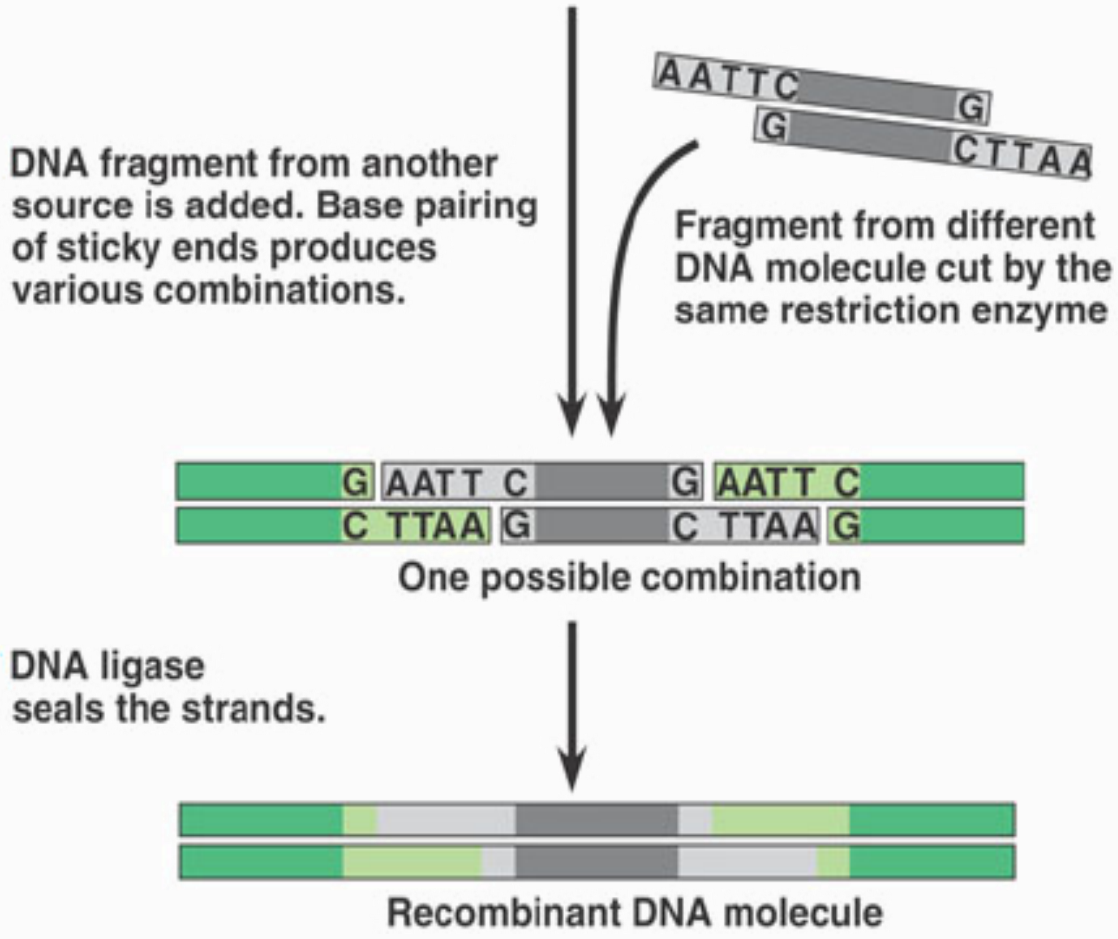
Blunt & Sticky ends



AluI and HaeIII produce blunt ends

BamHI HindIII and EcoRI produce "sticky" ends

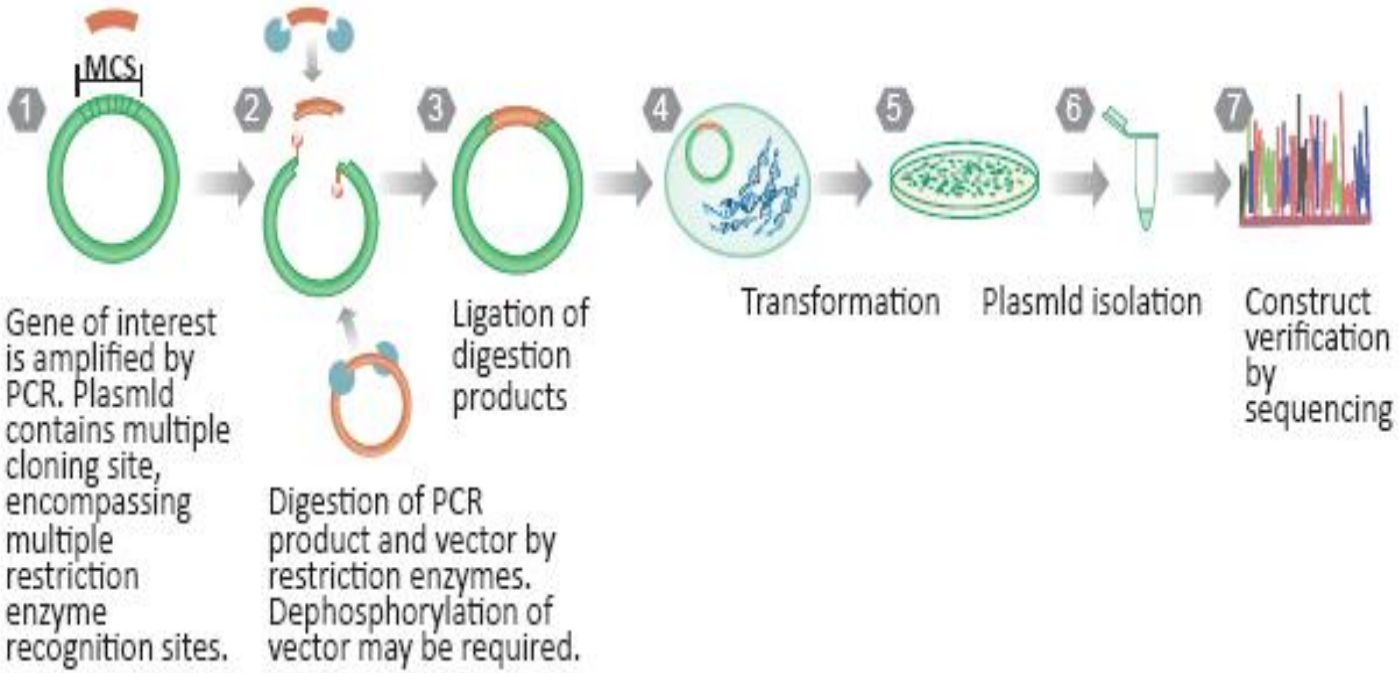
Pasting DNA



- **Complementary ends (sticky ends) H-bond**
- **Ligase forms phosphodiester bond to seal strands together.**

Dolly: The Cloning of a Sheep

- **Dolly** the sheep was successfully cloned in Britain in **1996** by the scientist "Ian Wilmut" and was put down in February **2003** after developing a **lung infection and arthritis**.
- Dolly was a **genetic copy** of the Finn Dorset ewe.
- Her birth, more than 10 years ago showed that nuclei from specialized adult cells can be reprogrammed into all the cells of an organism.
- The technique that led to Dolly is called
- **somatic cell nuclear transfer** and has
- remained essentially unchanged over
- the last decade.


MCS
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Gene of interest is amplified by PCR. Plasmid contains multiple cloning site, encompassing multiple restriction enzyme recognition sites.
Digestion of PCR product and vector by restriction enzymes. Dephosphorylation of vector may be required.
Ligation of digestion products
Transformation
Plasmid isolation
Construct verification by sequencing